

🔗 ENCODE CLUB × CIRCLE · PROGRAMMABLE MONEY HACKATHON

Aura DCA

An autonomous DCA agent that lets **Claude drive strategy** — while **code owns every number that touches money**. Built on Arc Network.

🌐 aura-dca.xyz

🔗 Arc Testnet

🧠 Decisions by Claude

🔵 Circle Swap Kit

01 · THE PROBLEM

You can't let an LLM be the final authority on money.

Give it the keys and one hallucinated number drains the wallet. Take away all its agency and it's just a cron script with extra steps.

And the moment the money isn't yours, it must be **fair** as well as **safe** — every user's schedule honoured, every user's funds ring-fenced, every allocation reconstructable — while still settling efficiently on-chain.

Claude recommends. Code decides.

🧠 Claude — the agent

Recommends an amount + reasoning.

Reads balance, day count, budget, its own history.

```
{ proceed, amountUsdc, reasoning }
```

Never trusted with the final number.

🔒 clampDecision() — the code

Decides the amount actually swapped.

Re-derives the cap from hard guardrails: max/day, reserve, dust, campaign budget.

Sole authority. Pure function. Unit-tested.

Records boundBy — which rule bound the result.

03 · THE FLAGSHIP CLAIM

Many users. **One swap.** Fair settlement.

- 1 Each wallet sets its **own** cadence, amount, and caps.
- 2 Every run, the agent executes the **sum as a single swap** per token.
- 3 Proceeds distributed **pro-rata by contribution** — remainder assigned deterministically so the books always close.

One transaction serves everyone. **Nobody subsidises anyone.**

Two wallets. One swap. Split to the last unit.

2.00 USDC → 1.793953 EURC · tx 0xd8a19f...1527

WALLET	CONTRIBUTED	RECEIVED	SHARE
0xdd6045a6...	1.000000 USDC	0.896977 EURC	50.0%
0xfc337ba1...	1.000000 USDC	0.896976 EURC	50.0%

0.896977 + 0.896976 = 1.793953 — the books close exactly.
Executed on the hourly cron, **unsupervised**.

Both halves of Circle's wallet stack.

The agent

Developer-Controlled Wallet — Circle custodies the signing key server-side. No raw private key to leak in a CI secret.

The user

Sign in with Google → a real, non-custodial **User-Controlled Wallet on Arc**. MPC custody, PIN recovery — no MetaMask, no seed phrase.

Every state change is authorised by the user's **own signature**. The agent executes the strategy — it can never invent a user's consent.

The ledger is **tamper-evident on-chain**.

After each run the agent records `keccak256(data/ledger.json)` in **AuraAttestation.sol** — a purpose-built Arc Testnet contract only its wallet may write to.

14+

attestations written by the cron — no human in the loop

0

funds held — a bug there can't move a token

verify-attest

anyone recomputes the hash, read-only

The agent knew when **not** to act.

USDC → cirBTC returned "*No route available*" across 14 distinct days — a liquidity outage on Arc Testnet, **not a bug in the agent** (proven isolated by check-routes).

It recognised the failure as **structural, not transient**, wrote that to its own reflections, cut its probe frequency, and **withheld spend to preserve capital** — unsupervised. Pick EURC and buys settle live today; leave it on cirBTC and the agent rides out the outage. Both run side by side.

The agent pays for its **own inputs**.

Beyond *spending* USDC to accumulate, it can **pay per call** over **x402** (HTTP 402): a metered "market brief" unlocked by a signed USDC authorization — EIP-3009 `transferWithAuthorization`.

```
npm run x402-demo # one command · no keys · no setup
```

Honest scope: signed & cryptographically **verified**, settlement gated off — an additive module (12 tests) that never touches the DCA pipeline.

Tested code owns the money path.

TypeScript · tsx

Anthropic Claude · tool-use + zod

Circle Swap Kit

Dev-Controlled Wallets

User-Controlled Wallets

Arc Testnet (EVM L1)

Solidity · AuraAttestation

GitHub Actions cron

Vercel serverless

Vitest

65

unit tests on the safety-critical paths

2

switches guard live trading: DRY_RUN + LIVE_TRADING_ENABLED

0

servers to host — self-hosted on free CI

Proven, not asserted.

- ◆ The **hard claim is proven on-chain** — many users, one swap, split to the last unit. Twice, unsupervised.
- ◆ **Real, verifiable execution** — real swaps, real per-user distributions, green CI anyone can inspect. Not a demo video.
- ◆ **Safety architecture** — LLM recommends / code decides, plus a two-switch live gate.
- ◆ **It audits itself on-chain** and reports to Telegram every run — genuinely autonomous.

TWO ORBITS, MEETING AT EVERY DECISION

Judgment, on rails.

Claude reasons about the market and proposes. The code re-derives the limit and owns the number that actually gets spent. That intersection is where the safety of an LLM-driven money agent lives.

 aura-dca.xyz

 github.com/thanhphuc85/AuraDCA

 [Verify the swap on ArcScan](#)